

Detection of dengue virus in *Aedes aegypti* during an urban epidemic in Iquitos, Peru (December 2010 to March 2011): results and tables

Anna B. Kawiecki ORCID: 0000-0002-0499-2612

Load libraries

```
# Handles relative file paths
library(here)

# Grammar of tables for publication-quality outputs
library(gt)

# Simplifies working with dates and times
library(lubridate)

# Spatial data handling and simple features for mapping
library(sf)

# Core tidyverse packages for data manipulation and visualization
library(tidyverse)
```

Read in data

```
# Household-level data
m.ae.pcr <- readRDS(here("analysis", "data", "raw_data", "m.ae.pcr.rds") )
m.ae.mid.pcr <- readRDS(here("analysis", "data", "derived_data",
                             "household_level_data", "m.ae.mid.pcr.rds"))
m.surv <- readRDS(here("analysis", "data", "derived_data",
                       "household_level_data", "m.surv.rds"))
m.h.surv <- readRDS(here("analysis", "data", "derived_data",
                         "household_level_data", "m.h.surv.rds"))
h.surv <- readRDS(here("analysis", "data", "derived_data", "household_level_data",
                       "h.surv.rds"))
```

RESULTS

Average number of surveys per week.

```
m.surv.means <- m.h.surv %>%
  st_drop_geometry() %>% # Remove geometry column
  filter(host == "mosquito") %>% # Keep mosquito data only
  group_by(epiweek) %>% # Group by epidemiological week
  summarise(n = n()) # Count surveys per week
```

```
summary(m.surv.means$n)           # Summary of weekly survey counts

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    175.0  261.8   337.0   328.0   369.5   498.0

l.surveys.week <- lm(n ~ 1, m.surv.means) # Fit intercept-only model

confint(l.surveys.week, level = 0.95)    # 95% CI for mean surveys/week

##              2.5 %    97.5 %
## (Intercept) 287.4761 368.5239
```

Average number of *Ae. aegypti* females collected per week.

```
# View summary statistics for Ae. aegypti females per survey-week
summary(m.surv$aa_female)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.0000  0.0000  0.0000  0.5266  0.0000 127.0000

# Fit intercept-only linear model for female counts
l.aa_f <- lm(aa_female ~ 1, m.surv)

# Compute 95% confidence interval for mean female count
confint(l.aa_f, level = 0.95)

##              2.5 %    97.5 %
## (Intercept) 0.4710774 0.5821964
```

Number of surveys with 0 *Ae. aegypti* females collected per week.

```
m.surv.0 <- m.surv %>% filter(aa_female == 0) # Filter for surveys with zero females

(nrow(m.surv.0)/nrow(m.surv))*100 # Count number of surveys with)

## [1] 76.88988
```

Number of surveys with more than 5 *Ae. aegypti* females collected

```
m.surv.high_aaf <- m.surv |>
  filter(aa_female > 5)

(nrow(m.surv.high_aaf)/nrow(m.surv))*100 # 1.354167

## [1] 1.354167

# Only 1.35 % of surveys had more than 5 female Ae.aegypti
```

Average number of *Ae. aegypti* females collected per week in houses with at least 1 infected female.

```
m.surv.positive <- m.surv %>% filter(denv.house == "positive")
# View summary statistics for Ae. aegypti females per survey-week
summary(m.surv.positive$aa_female)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   1.000   2.000   5.187   5.000  127.000
```

Average number of *Ae. aegypti* females collected per week in houses with no infected females.

```
m.surv.negative <- m.surv %>% filter(denv.house == "negative")
# View summary statistics for Ae. aegypti females per survey-week
summary(m.surv.negative$aa_female)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   1.000   1.000   2.074   2.000   63.000
```

Average number of *Ae. aegypti* females collected per week in houses with any females.

```
m.surv.any <- m.surv %>% filter(aa_female > 0)
# View summary statistics for Ae. aegypti females per survey-week
summary(m.surv.any$aa_female)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000   1.000   1.000   2.279   2.000  127.000
```

How many surveys collected females that were infected?

```
# Filter for surveys where DENV-positive Ae. aegypti were detected
m.house.infected <- m.ae.pcr %>%
  filter(denv.house == "positive") %>%
  # Flag duplicate locations
  mutate(dup = duplicated(location_code))

# Count number of surveys (rows)
nrow(m.house.infected)
```

```
## [1] 75
```

Surveys where females were tested

```
table(m.surv$ae.f.survey)
```

```
##
##      No Ae.ae females collected      No DENV test on Ae.ae females
##              5167              308
## DENV negative (all Ae.ae females)  DENV positive (>=1 Ae.ae female)
##              1170              75
```

```
nrow(m.surv |> filter(ae.f.survey == "DENV positive (>=1 Ae.ae female)"))
```

```
## [1] 75
```

```
# 75
```

```
nrow(m.surv |> filter(ae.f.survey == "DENV negative (all Ae.ae females)"))
```

```
## [1] 1170
```

```
# 1170
nrow(m.surv |> filter(ae.f.survey == "DENV positive (>=1 Ae.ae female)")) +
  nrow(m.surv |> filter(ae.f.survey == "DENV negative (all Ae.ae females)"))

## [1] 1245
# 1245
```

In surveys where females were tested: distribution of adult females and positive surveys

```
# Surveys where females were tested
denv.house.num <- m.surv |> filter(!is.na(denv.house.num))

# Proportion of surveys where none of the tested females were positive
nrow(denv.house.num |> filter(denv.house.num == 0))/nrow(denv.house.num) # 0.94

## [1] 0.939759

# Proportion of surveys where at least one of the tested females were positive
nrow(denv.house.num |> filter(denv.house.num == 1))/nrow(denv.house.num) # 0.06

## [1] 0.06024096

# Proportion of surveys where females were tested with only 1 female collected
nrow(denv.house.num |> filter(aa_female == 1))/nrow(denv.house.num) # 0.58

## [1] 0.5823293

# Proportion of surveys where females were tested with >5 females collected
nrow(denv.house.num |> filter(aa_female > 5))/nrow(denv.house.num) # 0.057

## [1] 0.05702811

sum(denv.house.num$n) # 2795

## [1] 2795

sum(denv.house.num$n.pos.mosq) # 128

## [1] 128

sum(denv.house.num$n.pos.mosq)/sum(denv.house.num$n) # 0.04579

## [1] 0.04579606

summary(denv.house.num$aa_female)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.000  1.000   1.000   2.262  2.000 127.000
#  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#  1.000  1.000   1.000   2.262  2.000 127.000
```

How many households had DENV-infected females?

```
# Number of unique household IDs where DENV-positive Ae. aegypti were detected
length(unique(m.house.infected$location_code)) #72
```

```
## [1] 72
```

How many households had infected females found in multiple surveys?

```
# Subset for households with duplicated positive detections
m.house.infected.multi <- m.house.infected[which(m.house.infected$dup == TRUE),] %>%
  st_drop_geometry() # Drop spatial geometry

# Count unique households with repeated positive detections
length(unique(m.house.infected.multi$location_code))
```

```
## [1] 2
```

Number of days between household surveys in households where infected females were found repeatedly:

```
# Get survey dates for the 1st household with repeated infection
m.house.infected.multi.1 <- m.house.infected %>%
  filter(location_code == m.house.infected.multi$location_code[[1]] ) %>%
  dplyr::select("date.survey") %>%
  st_drop_geometry()

# View the selected survey dates
m.house.infected.multi.1

## # A tibble: 2 x 1
##   date.survey
## * <date>
## 1 2011-03-11
## 2 2011-03-15

# Calculate the number of days between two surveys (hardcoded)
days_diff.1 <- as.numeric(difftime(ymd("2011-03-11"), ymd("2011-03-15"), units = "days"))

# View result
days_diff.1 # 4
```

```
## [1] -4
```

```
# Get survey dates for the 2nd household with repeated infection
m.house.infected.multi.2 <- m.house.infected %>%
  filter(location_code == m.house.infected.multi$location_code[[2]] ) %>%
  dplyr::select("date.survey") %>%
  st_drop_geometry()

# View the selected survey dates
m.house.infected.multi.2
```

```
## # A tibble: 3 x 1
##   date.survey
## * <date>
## 1 2010-12-14
## 2 2011-01-07
```

```
## 3 2011-01-14
# Calculate days between two surveys
days_diff.2 <- as.numeric(difftime(ymd("2010-12-14"), ymd("2011-01-14"), units = "days"))

# View result
days_diff.2 # 31

## [1] -31
```

Average percent of infected females in households with infected females

```
# Average percent of infected females in households with one or more infected females
mean(m.house.infected$perc.pos) #68.07393

## [1] 68.07393
```

Number of the DENV positive females that were not dissected

```
m.ae.mid.pcr.pos <- m.ae.mid.pcr %>%
  st_drop_geometry() %>% # Remove spatial geometry
  filter(!is.na(mosquito_id)) %>% # Exclude records with missing mosquito IDs
  filter(pcr == "positive") %>% # Keep only PCR-positive results
  group_by(n.body.parts.tested, # Group by number of body parts tested,
            pcr.positive.body, # PCR-positive body part,
            date) %>% # and date of survey
  summarise(n = n()) %>% # Count number of records per group
  filter(n.body.parts.tested == 1 | # Keep if only 1 body part tested,
         is.na(n.body.parts.tested)) # or if the entire body was tested

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by n.body.parts.tested, pcr.positive.body,
## and date.
## i Output is grouped by n.body.parts.tested and pcr.positive.body.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(n.body.parts.tested, pcr.positive.body, date))` for
## per-operation grouping (`?dplyr::dplyr_by`) instead.

# Calculate how many positive samples had abdomen tested separately (out of 128 total)
128 - sum(m.ae.mid.pcr.pos$n)

## [1] 123
```

% females with detectable DENV in the head-thorax-legs-wings

```
# Summarize PCR-positive mosquitoes tested on 2 body parts
m.ae.pcr.mid.sum <- m.ae.mid.pcr %>%
  st_drop_geometry() %>% # Drop spatial geometry
  filter(pcr == "positive" & # Keep PCR-positive cases
         n.body.parts.tested == 2) %>% # Only where 2 body parts were tested
  filter(pcr.positive.body != # Exclude where whole mosquito tested
         "abdomen or thorax") %>%
  mutate(dissemination.evidence = case_when( # Classify dissemination
    pcr.positive.body == "abdomen" ~
      "No evidence of dissemination",
```

```

  T ~ "Evidence of dissemination"
)) %>%
group_by(dissemination.evidence) %>%      # Group by dissemination class
summarise(n = n())                        # Count observations per group

# Proportion with evidence of dissemination
m.ae.pcr.mid.sum[
  which(m.ae.pcr.mid.sum$dissemination.evidence ==
        "Evidence of dissemination"),
][["n"]] / sum(m.ae.pcr.mid.sum$n)

## [1] 0.804878

```

TABLES

Table 1: Summary of adult mosquito surveys

```

table1 <- m.ae.pcr %>%
  st_drop_geometry() %>%      # Remove geometry column
  group_by(result, ae.f.survey) %>%      # Group by survey outcome
  summarise(n.houses = n(),      # Count of houses per group
            n.aa_f = sum(aa_female),    # Total female mosquitoes
            n.aa_f.test = sum(n),      # Number tested by PCR
            n.aa_f.denv = sum(n.pos.mosq)) %>% # Number DENV-positive
  ungroup() %>%
  mutate(n.total.surveys = sum(n.houses),      # Total surveys = 9405
         n.total.worked.surveys =
           sum(n.houses[result != "No survey"]), # Successful surveys = 6720
         n.total.worked.aa_f.surveys =
           sum(n.houses[ae.f.survey !=
                        "No Ae.ae females collected"]), # Houses where females were collected
         n.total.collected.aa_f =
           sum(n.aa_f, na.rm = TRUE),      # Collected females
         n.total.tested.aa_f =
           sum(n.aa_f.test, na.rm = TRUE),  # Tested females
         n.total.positive.aa_f =
           sum(n.aa_f.denv, na.rm = TRUE)) %>% # Positive females
  dplyr::select(-c("result", "ae.f.survey",
                   "n.houses", "n.aa_f",
                   "n.aa_f.test", "n.aa_f.denv")) %>% # Drop raw cols
  gather() %>% distinct() %>%      # Reshape, remove duplicates
  mutate(statistic = case_when(
    key == "n.total.surveys" ~ "Adult surveys",
    key == "n.total.worked.surveys" ~
      "Successful adult surveys",
    key == "n.total.worked.aa_f.surveys" ~
      "Adult surveys with female collections",
    key == "n.total.collected.aa_f" ~
      "Female adults collected",
    key == "n.total.tested.aa_f" ~
      "Female adults tested by PCR",
    key == "n.total.positive.aa_f" ~
      "Female adults DENV positive by PCR")) %>%

```

Table 1: Summary of adult mosquito surveys

	Number (%)
Adult surveys	
Adult surveys	9405
Successful adult surveys	6720 (71.5 %)
Adult surveys with female collections	1553 (16.5 %)
Female adults	
Female adults collected	3539
Female adults tested by PCR	2795 (79 %)
Female adults DENV positive by PCR	128 (4.6 %)

```
mutate(group = case_when(
  str_detect(statistic, "survey") == TRUE ~ "Adult surveys",
  T ~ "Female adults" # All others go to this group
)) %>%
group_by(group) %>% #
mutate(
  perc = case_when(
    group == "Adult surveys" ~ value / 9405 * 100,
    group != "Adult surveys" &
      statistic != "Female adults DENV positive by PCR" ~
        value / 3539 * 100,
    T ~ value / 2795 * 100)) %>%
mutate(perc = round(perc, digits = 1)) %>% # Round percentages
mutate(perc = gsub(100, "", perc)) %>% # Remove any 100 string
mutate(value = case_when( # Combine value and percent
  perc != "" ~ paste(value, "(", perc, "%)", ""),
  T ~ as.character(value)
)) %>%
rename('Number (%)' = value) %>% # Rename final column
dplyr::select(-c("key", "perc")) %>% # Drop unneeded columns
group_by(group) %>%
gt(rowname_col = "statistic") %>% # Format for display table
gt::tab_header(
  title = md("Table 1: Summary of adult mosquito surveys")
)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by result and ae.f.survey.
## i Output is grouped by result.
## i Use `summarise(groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(result, ae.f.survey))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
table1 # Print table
```

```
# Save as CSV for external use
write.csv(table1, here("analysis", "outputs", "tables", "table1.csv"))
```

Table 2: Host species identification

	Number (%)
Human	245 (98 %)
Avian	2 (0.8 %)
Dog	2 (0.8 %)
Cat	1 (0.4 %)

Table 2: Vertebrate host species identified from *Ae. aegypti* blood meals collected in Iquitos, December 2010- January 2011

```
# Create a data frame with species and counts
table2 <- data.frame(Species= c("Human", "Avian","Dog","Cat"),
                     Number= c(245,2,2,1)) %>%

  mutate(
    perc = Number / sum(Number) * 100 ) %>%      # Calculate percentages
  mutate(
    value = paste(Number, "(", perc, "%", ")") %>% # Combine count & percent
  rename('Number (%)' = value) %>%              # Rename final column
  dplyr::select(-c("Number", "perc")) %>%      # Drop intermediate columns
  gt(rowname_col = "Species") |>               # Create formatted gt table
  tab_header(
    title = md("Table 2: Host species identification") # Add table title
  )

table2 # Display the table

# Save table as CSV
write.csv(table2, here("analysis", "outputs", "tables", "table2.csv"))
```

Table 3: Adult mosquito surveys outcomes by surveillance strategy

```
table3 <- m.surv %>%
  # Remove geometry column
  st_drop_geometry() %>%
  # there are 2 females that weren't tested, so we have to calculate incidence only from those that were
  group_by(pos.case.contact) %>%
  summarise(
    # Number of surveys (houses worked)
    n.surveys= n(),
    # Total number Ae. aegypti collected (males + females)
    n.aa.total= sum(aa_total, na.rm= TRUE),
    # Number of Ae. aegypti females collected
    n.aa.f= sum(aa_female, na.rm= TRUE),
    # Number of Ae. aegypti females tested by PCR
    n.aa.f.test= sum(n, na.rm= TRUE),
    # Number of Ae. aegypti females tested by PCR that were DENV positive
    n.aa.f.denv = sum(n.pos.mosq, na.rm = TRUE),
    # Average number of total Ae. aegypti per house
    avg.aa.total= mean(aa_total, na.rm= TRUE),
    # Average number of Ae. aegypti females per house
```

```

    avg.aa.f= mean(aa_female, na.rm= TRUE),
  ) %>%
  # Select relevant columns for table
  dplyr::select(c("pos.case.contact", "n.surveys", "n.aa.f.test", "n.aa.f.denv", "avg.aa.f")) %>%
  # Calculate DENV % prevalence
  mutate(perc.aa.f.denv= (n.aa.f.denv/n.aa.f.test)*100) %>%
  # Rename surveillance strategy labels
  mutate(pos.case.contact= case_when(
    pos.case.contact=="other surveillance" ~ "Other surveillance",
    T~ "Positive case-contact"
  )) %>%
  # Rename columns for clarity
  rename('Average number of females/survey'=avg.aa.f,
    'Total number of surveys'=n.surveys,
    'Total number of females tested'=n.aa.f.test,
    'Total number of DENV positive females'=n.aa.f.denv,
    'Vector DENV prevalence (%)'=perc.aa.f.denv,
    'Surveillance strategy'=pos.case.contact ) %>%
  # Convert to long format for table display
  pivot_longer(
    cols = -'Surveillance strategy',
    names_to = "variable",
    values_to = "Value"
  ) %>%
  # Format numeric values to 1 decimal
  mutate(across(where(is.numeric), ~ round(.x, 2))) %>%
  # Keep only necessary columns
  dplyr::select(variable, 'Surveillance strategy', Value) %>%
  # Order variables logically
  mutate(variable= factor(variable,
    levels = c( 'Total number of surveys',
      'Total number of females',
      'Total number of females tested',
      'Total number of DENV positive females',
      'Average number of females/survey',
      'Vector DENV prevalence (%)')
  )) %>%
  # Arrange rows by variable order
  arrange(variable) %>%
  # Group by variable for table formatting
  # group_by(variable) %>%
  # # Create gt summary table
  gt(rowname_col = "variable") %>%
  # # Add table title
  gt::tab_header(
    title = md("Table 3: Adult mosquito surveys outcomes by surveillance strategy")
  )
)

table3

# Save table to CSV
write.csv(table3,here("analysis", "outputs", "tables", "table3.csv"))

```

Table 3: Adult mosquito surveys outcomes by surveillance strategy

	Surveillance strategy	Value
Total number of surveys	Other surveillance	6541.00
Total number of surveys	Positive case-contact	179.00
Total number of females tested	Other surveillance	2650.00
Total number of females tested	Positive case-contact	145.00
Total number of DENV positive females	Other surveillance	123.00
Total number of DENV positive females	Positive case-contact	5.00
Average number of females/survey	Other surveillance	0.51
Average number of females/survey	Positive case-contact	0.97
Vector DENV prevalence (%)	Other surveillance	4.64
Vector DENV prevalence (%)	Positive case-contact	3.45